

9.1 Implementation and Performance Analysis of LMS and nmls

Program:

```
clc; clear

x=randn(1,1000);
d=filter([0.8 0.5 0.3],1,x);

e1=d.exp(-0.01(1:1000));
e2=d.exp(-0.03(1:1000));

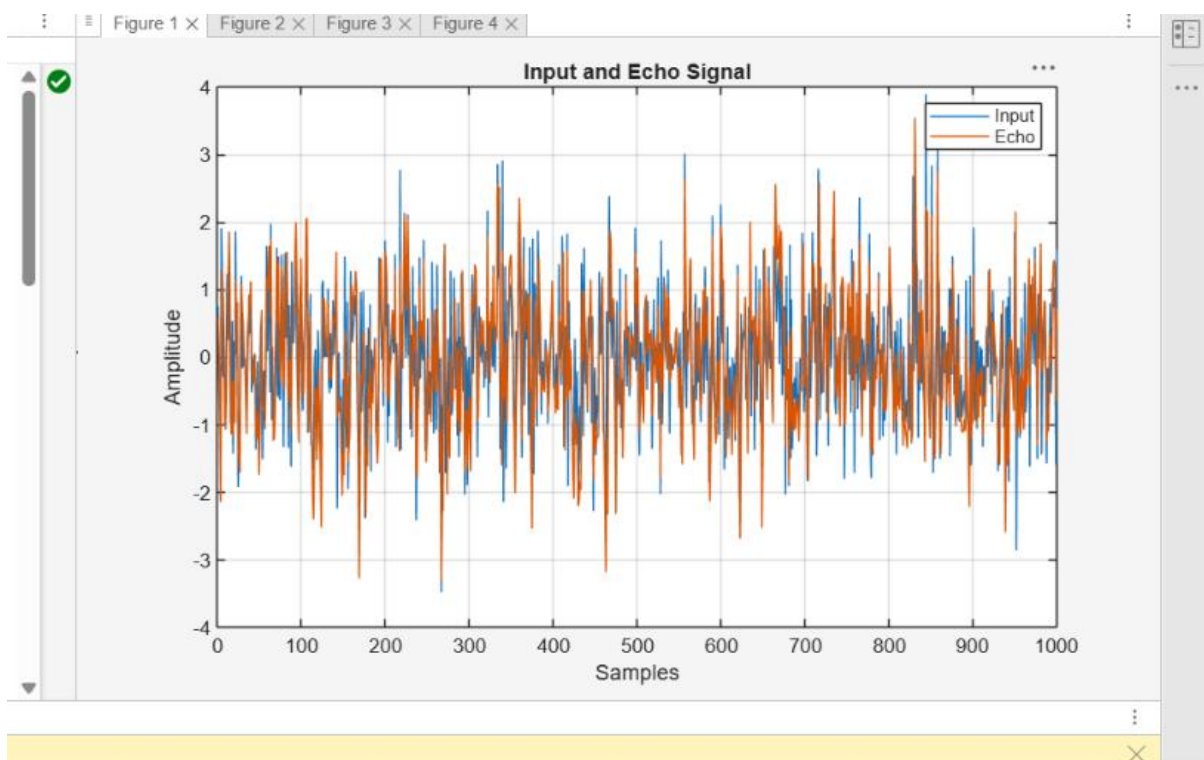
figure

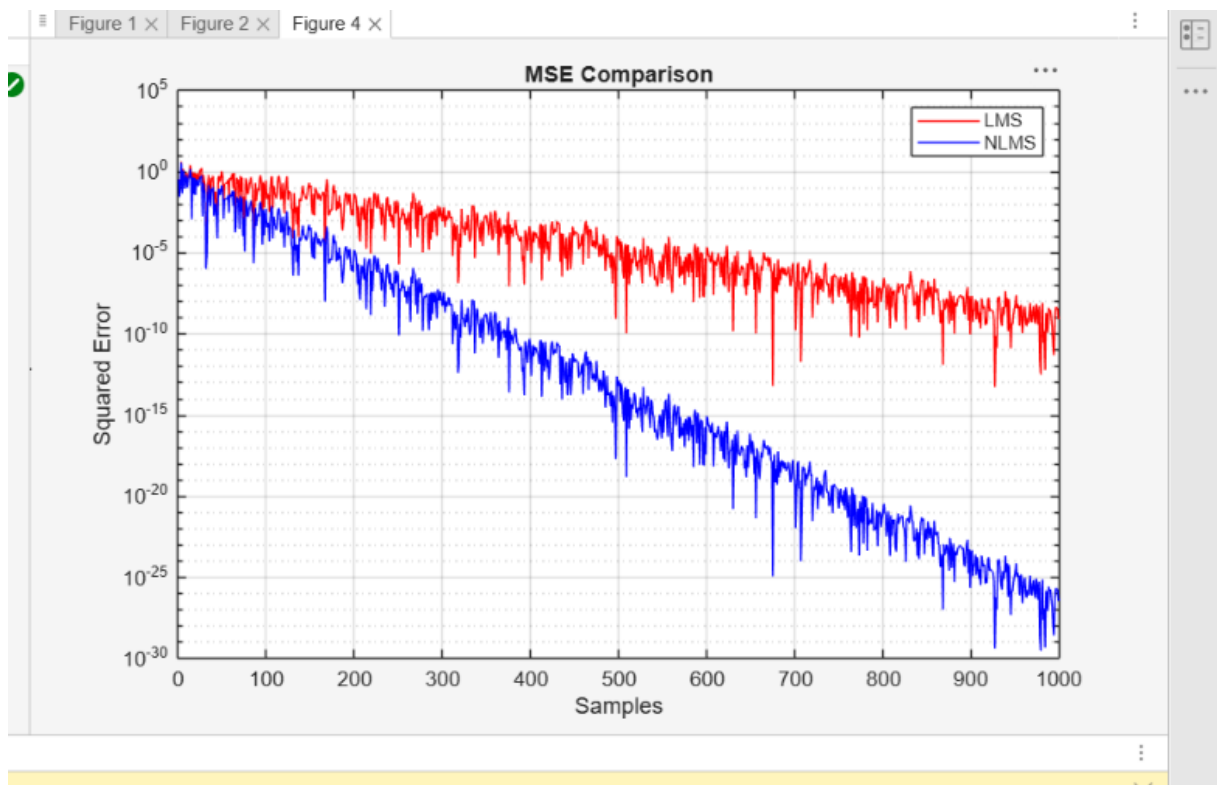
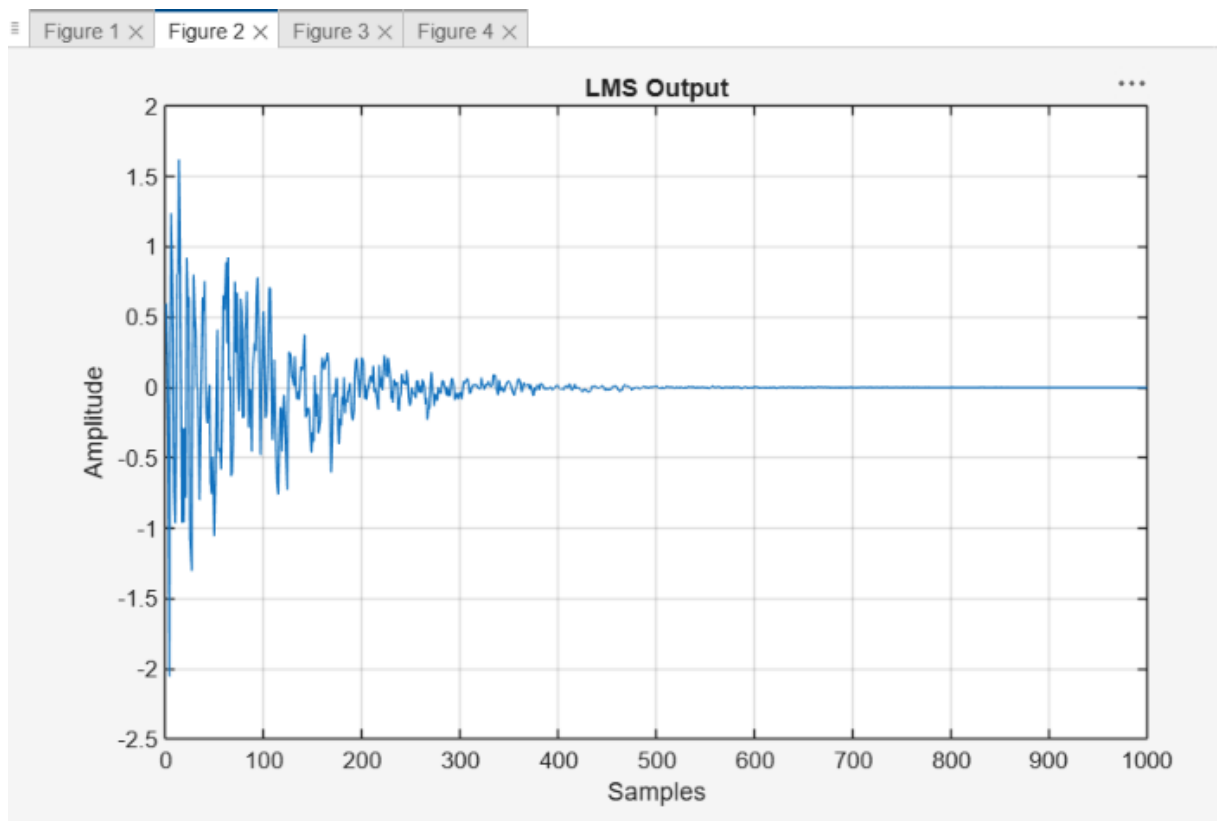
plot(x)
hold on
plot(d)
grid on
legend('Input','Echo')
title('Input and Echo Signal')

figure
plot(e1)
grid on
title('LMS Output')
```

```
figure
plot(e2)
grid on
title('NLMS Output')
```

```
figure
semilogy(e1.^2,'r')
hold on
semilogy(e2.^2,'b')
grid on
legend('LMS','NLMS')
title('MSE Comparison')
```





10.1 Implementation of Spectral Subtraction for Speech Noise Reduction

Program:

```
clc;
clear;
close all;

Fs = 8000;
t = 0:1/Fs:3;

% Clean signal
x = sin(2*pi*200*t) + 0.5*sin(2*pi*400*t);

% Noisy signal
xn = x + 0.3*randn(size(x));

% FFT of noisy signal
Y = fft(xn);

% Estimate noise
N = fft(0.3*randn(size(x)));

% Spectral subtraction
mag = max(abs(Y) - abs(N), 0);
xe = real(ifft(mag .* exp(1j*angle(Y))));

% Signals
figure;
```

```
subplot(3,1,1);  
plot(t,x);  
title('Clean Speech');  
xlabel('Time (s)');  
grid on;
```

```
subplot(3,1,2);  
plot(t,xn);  
title('Noisy Speech');  
xlabel('Time (s)');  
grid on;
```

```
subplot(3,1,3);  
plot(t,x_e);  
title('Enhanced Speech');  
xlabel('Time (s)');  
grid on;
```

% Spectrograms

```
figure;  
subplot(3,1,1);  
spectrogram(x,256,128,256,Fs,'yaxis');  
title('Clean Spectrogram');
```

```
subplot(3,1,2);  
spectrogram(xn,256,128,256,Fs,'yaxis');  
title('Noisy Spectrogram');
```

```
subplot(3,1,3);
```

```
spectrogram(xe,256,128,256,Fs,'yaxis');  
title('Enhanced Spectrogram');
```

```
% Magnitude Spectra  
figure;  
subplot(3,1,1);  
plot(abs(fft(x)));  
title('Clean Spectrum');  
xlabel('Frequency');  
grid on;
```

```
subplot(3,1,2);  
plot(abs(fft(xn)));  
title('Noisy Spectrum');  
xlabel('Frequency');  
grid on;
```

```
subplot(3,1,3);  
plot(abs(fft(xe)));  
title('Enhanced Spectrum');  
xlabel('Frequency');  
grid on;
```

